



February 16, 2016

Ms. Maria Pallante
Register of Copyrights
United States Copyright Office
101 Independence Ave., S.E.
Washington, D.C. 20559

Re: **Software-Enabled Consumer Products Study; Notice and Request for Public Comments - Docket No. 2015-6**

Dear Ms. Pallante:

BSA | The Software Alliance (“BSA”) appreciates this opportunity to respond to the Copyright Office’s Request for Public Comments on the “role of copyright law with respect to software-enabled consumer products.”¹ As the leading advocate for the global software industry in the United States and around the world, BSA is uniquely positioned to provide insights from the very companies that are driving the current wave of software-enabled innovation. BSA members are among the world’s most entrepreneurial companies, creating software solutions that spark the economy and improve modern life.² In short, BSA’s member companies breathe life into the “everyday products” that are the focus of this study by providing the software that makes these devices desirable.

This study’s focus on how “the application of copyright law to software in everyday products enables or frustrates innovation” comes at an interesting time.³ Of course, the existence of software in consumer devices is hardly a new phenomenon; a huge number of “everyday products,” including microwave ovens and handheld calculators, have since the early-1970s featured embedded software. However, U.S. software companies are on the leading edge of developing new network-connected technologies that are giving rise to what economists from the Massachusetts Institute of Technology have deemed the “Second Machine Age,” a period in which software-enabled innovations will be “as important and transformational to society and the economy as the steam engine.”⁴

Given the rapid pace of innovation that is occurring in network-connected intelligent devices, it is unclear what “problems” this study may seek to address through changes to the copyright law. The Office cautions the study “is not intended to examine or address more general questions about software and copyright protection.” However, we are concerned that the Office’s questions may suggest a bias in favor of changing the law to create a *sui generis* (and perhaps diminished)

¹ U.S. Copyright Office, Library of Congress, *Software-Enabled Consumer Products Study; Notice and Request for Comments*, 80 Fed. Reg. 77668-71 (Dec. 15, 2015) [hereinafter *Software-Enabled Study RFC*].

² BSA’s members include: Adobe, ANSYS, Apple, Autodesk, Bentley Systems, CA Technologies, CNC/Mastercam, DataStax, Dell, IBM, Intuit, Microsoft, Minitab, Oracle, Salesforce, SAS Institute, Siemens PLM Software, Symantec, Tekla, The MathWorks, Trend Micro and Workday.

³ *Software-Enabled Study* at 77668.

⁴ Erik Brynjolfsson & Andrew McAfee, *The Second Machine Age: Work, Progress and Prosperity in a Time of Brilliant Technologies* 9 (W. W. Norton & Co. 2014).

form of copyright protection for software embedded in “everyday products.” Absent a strong evidentiary record that copyright law is – in fact rather than in theory -- having unintended, negative effects on the development of software-enabled products, such a proposal seems very much like “solution” in search of a problem.

Furthermore, the entire software industry is highly dynamic and interconnected, which makes it virtually impossible to draw any principled distinction between embedded and non-embedded software. Indeed, the software on phones, tablets, and other personal devices interact directly with, and borrow substantial functionality from, enterprise-grade software running in centralized servers. As one example, voice recognition technology in many handsets depends heavily on the shared processing of the audio signal between the handset and a central voice processing server. It is extremely likely that modifications to the law of copyright as it relates to everyday products would have material, and potentially unpredictable, unintended impacts on other areas of the software industry. In the absence of any evidence of consumer injury, there is no justification for making changes to copyright law as it relates to embedded software.

Whether Copyright Protection for Embedded Software is Spurring or Frustrating Innovation

As the Copyright Office’s Request for Comments observes, Congress extended copyright protection to software in order to promote innovation and creativity.⁵ Although today’s software industry looks far different from the nascent industry that existed when Congress expressly listed “computer programs” as copyrightable subject matter in 1980, the potential for software to become integral to the daily lives of ordinary people was apparent even then. Indeed, in its recommendation that software receive the full scope of copyright protection afforded to other forms of creative works, the National Commission on New Technological Uses of Copyrighted Works (“CONTU”) recognized that:

“The possibilities provided by the technology are virtually limitless. They are dependent only on the ingenuity employed in developing the programs that enable humans to communicate their ideas to one another through the intermediation of the machine and on the willingness of creators of such works to disseminate them at reasonable prices. In considering the quality of life in this country, failing to weigh the positive contributions of computers and the programs with which they are used would indeed be a mistake.”⁶

By any measure, CONTU’s vision, as ratified by Congress in 1980, has been realized. The existing U.S. copyright framework for software has given rise to the most innovative and diverse software industry in the world. With nine of the top ten software companies by revenue residing in the United States,⁷ the U.S. software industry is responsible for many of the technological advances that have reshaped virtually every facet of our lives over the past two decades. To name just a few examples, U.S. software companies are helping to: change the face of education by shrinking textbooks to tablets and connecting classrooms across continents; deliver cutting-edge health and fitness innovation; make our infrastructure safer, smarter and more collaborative; and improve the efficiency of our transit systems.⁸

⁵ See Software-Enabled Study RFC at 77669 (“Though the 1976 Act did not expressly list computer programs as copyrightable subject matter, *the Act’s legislative history makes it evident that Congress intended for them to be protected by copyright law . . .*”) (emphasis added).

⁶ National Commission on New Technological Uses of Copyrighted Works, Final Report of the National Commission on New Technological Uses of Copyrighted Works 26 (1978) (“CONTU Report”).

⁷ Shobhit Set, *World’s Top 10 Software Companies*, Investopedia.com (Jan. 8, 2015), at <http://www.investopedia.com/articles/personal-finance/121714/worlds-top-10-software-companies.asp>

⁸ See, BSA | The Software Alliance, *You Can Do Amazing Things With Software*, at <http://withsoftware.org/>.

There is no indication that this trend is slowing or will slow anytime soon. In fact, we are on the cusp of an era of even greater software-driven innovation, due in large measure to strong and comprehensive copyright protection for software, that is being enabled by the convergence of Internet-connected devices, cloud computing capabilities, and data analytics. The “Internet of Things” (“IoT”) describes the growing network of “smart” devices that are software-enabled and Internet-connected. These software-powered devices can collect and exchange data and add intelligence to improve efficiency, bolster performance, and add functionality. Unlike Internet-connected devices that have been around for decades, software now enables “devices of all kinds – including smartphones, wearables, appliances, medical equipment, and vehicles – to connect with the Internet and each other to create, share and analyze information, all without human intervention.”⁹

Embedded software is at the very core of IoT innovations that are transforming industries and the manner in which consumers interact with their devices. Consumers now have access to a range of IoT products capable of improving “conservation, efficiency, productivity, public safety, health, education and more.”¹⁰ For instance, new automobiles are equipped with more than 10 million lines of software code to control sensors that generate up to 25 gigabytes of data per hour that can be processed to enable safety and crash avoidance systems, reduce congestion and route traffic more efficiently.¹¹ Both Apple and Microsoft have launched wearable devices with sensors that measure biometric data and provide unprecedented insights to help consumers optimize their workouts and live healthier lives.¹² And by 2020, analysts predict that there will be more than 50 billion IoT devices relied upon by consumers, governments and businesses.¹³

The Request for Public Comments asks whether “copyright law should distinguish between software embedded in ‘everyday products’ and other types of software.”¹⁴ In view of the ongoing transformations occurring in consumer technology and the rapid pace at which advances are occurring, the answer must be “no.” In the past 10 years, embedded software has turned our mobile phones into fully functional computers and connected our home thermostats and other electronic devices to the Internet to improve energy efficiency. And the pace of change is only accelerating. Over the next 10 years, embedded software may usher an era in which driverless cars are the norm and implantable technologies help to manage chronic diseases. Over time, these developments may raise public policy questions across a range of issues, such as privacy, security and public safety. But it strains credulity given the success in the market to date that the “application of copyright law to software in everyday products... [is frustrating] innovation.”

Whether Copyright Protection for Embedded Software Frustrates Interoperability

Consumers increasingly (and justifiably) demand flexibility in the ways they access, use, and enjoy network-connected devices – including in ways that were unimaginable only a few years ago. A well-functioning copyright system should therefore encourage rights holders to make use of emerging technologies to develop new products and services to satisfy consumer demand. When rights holders have the flexibility to experiment with and implement different business

⁹ Deloitte, *The Internet of Things Ecosystem: Unlocking the Business Value of Connected Devices* 5 (2014), at <http://www2.deloitte.com/content/dam/Deloitte/global/Documents/Technology-Media-Telecommunications/gx-tmt-lotecosystem.pdf>

¹⁰ Consumer Electronics Association, Comment to the Federal Trade Commission on Internet of Things, Project No. P135405 (June 10, 2013) 7.

¹¹ BSA | The Software Alliance, *What's the Big Deal With Data?* 15 (2015), at http://data.bsa.org/wp-content/uploads/2015/12/bsadatastudy_en.pdf

¹² See <https://www.microsoft.com/Microsoft-Band/en-us>; <http://www.apple.com/watch/>

¹³ See Valerio, Pablo, *Internet of Things: 50 Billion Is Only the Beginning*, EE Times, Feb. 28, 2014, http://www.eetimes.com/document.asp?doc_id=1321229

¹⁴ Software-Enabled Study at 77671.

models, market competition will work to the public's benefit. One metric for evaluating the health of a copyright system is the extent to which it supports myriad business models.

In examining whether copyright is enabling or frustrating interoperability with software-embedded devices, it's important to bear in mind that interoperability is largely a function of business strategy. Title 17 provides rights holders with the flexibility to develop software-embedded devices and platforms along a continuum of development and business models. Because consumers benefit from being able to select a model that best fits their needs, there is a strong public policy interest in preserving this type of vibrant market competition.

Marketplace needs and consumer demands for products that work together (interoperate) within larger information processing systems are in most instances addressed by developing industry standards. In those circumstances, interoperability is effectively promoted through standards setting efforts entered into voluntarily by industry. Such efforts have been underway for decades in many fields where key technologies included in the standard are protected by copyright or patents. IP laws have not been an obstacle; to the contrary, they have enabled creators and inventors to contribute their technologies to the standard knowing their rights are protected and their incentives to further innovate are promoted. IEEE 802.11 is an example of such a standard facilitating multi-vendor interoperability by providing a specification that enables "everyday products" to connect to wireless networks using a common protocol.

Like standards, innovative uses of application programming interfaces ("APIs") are also driving an era of increased interoperability. At their most basic, APIs provide an interface through which software applications are able to communicate and make requests of one another. Software developers are increasingly using APIs to ensure that their products are interoperable with a wide range of hardware and platform configurations. For instance, Autodesk recently launched Spark, an open 3D printing software platform with APIs that enable third party companies to build 3D printing capabilities directly into their products. By opening the Spark platform to 3D printing software developers and manufacturers, Autodesk is creating an ecosystem of interoperable 3D printing software and hardware solutions.

In the absence of clear evidence that copyright law is inhibiting consumers from accessing a wide variety of product offerings, we would caution against entertaining potential legislative changes to Title 17. Any such change could destabilize a legal environment that has supported innovation, fostered consumer choice, and encouraged interoperability over more than three decades. Weakening copyright protection for embedded software in an effort to encourage specific business outcomes -- whether in terms of licensing practices or design choices -- could distort or diminish the existing positive market developments and unnecessarily disrupt stable international norms for the protection of software under copyright.

Whether Copyright Protection for Embedded Software Frustrates Secondary Markets

The first sale doctrine has since the 1908 Supreme Court decision in *Bobbs-Merrill Co. v. Straus*¹⁵ played an important role in ensuring that copyright protection does not act as an inadvertent restraint on the alienation of physical property. By upholding "the distinction between ownership of the copyright and ownership of the material object" on which a copy of a work resides,¹⁶ the first sale doctrine allows the owner of a physical copy of a work to resell or

¹⁵ 210 U.S. § 339 (1908).

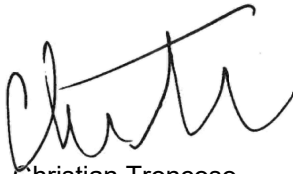
¹⁶ U.S. Copyright Office, *A Report of the Register of Copyrights Pursuant to § 104 of the Digital Millennium Copyright Act*, 87 (2001), available at <http://www.copyright.gov/reports/studies/dmca/sec-104-report-vol-1.pdf> ("Section 104 Report").

otherwise dispose of that copy without the copyright owner's consent.¹⁷ Because Section 109 applies without limitation to software that is "embodied in a machine or product and which cannot be copied during the ordinary operation of the machine or product,"¹⁸ the first sale doctrine appropriately supports secondary markets for products that contain embedded software.

Although the first sale doctrine does not apply to software obtained by license,¹⁹ there is remarkably little evidence that companies are structuring their licensing agreements to prevent the emergence of legitimate secondary markets for products with embedded software. In fact, the Department of Commerce's Internet Policy Task Force recently concluded that the risk of manufacturers using license restrictions on embedded software to prevent the development of secondary markets is largely theoretical.²⁰ Theoretical concerns and decades-old anecdotes are flimsy justification for changing the law. Absent clear, current evidence of specific problems that are not being addressed by the market, we believe it is premature to recommend legislative changes that could create unintended distortions in a quickly evolving market.

Thank you again for the opportunity to share our views on these important issues.

Sincerely,

A handwritten signature in black ink, appearing to read 'Christian Troncoso', with a stylized, cursive script.

Christian Troncoso
Director, Policy

¹⁷ See 17 U.S.C. § 109.

¹⁸ 17 U.S.C. § 109(b)(1)(B)(i).

¹⁹ 17 U.S.C. § 109(d).

²⁰ Department of Commerce, Internet Policy Task Force, *White Paper on Remixes, First Sale, and Statutory Damages*, 64 (2016) ("The Task Force did not hear evidence that licenses purporting to restrict a consumer's ability to resell have been used with respect to embedded software that operates a functional product, other than a computer or related equipment. Thus, the record before us does not establish that the kinds of consumer products identified above are currently sold subject to such licenses.")